

4(a)	any three from: 1 formation of new species from, pre-existing species / (common) ancestor ; A description 2 over, time / generations ; 3 mutations / changes in gene pools / changes in allele frequencies ; 4 change in, characteristics / phenotypes ; 5 natural selection ; 6 AVP ; e.g. genetic drift / allopatric / sympatric	3
4(b)(i)	A names for numbers throughout any three from: max two for similarities in both figures: 1 diverged into two groups, at the same time / 67 million years ago 2 <i>idea that</i> group 1 is on a different (evolutionary) branch to group(s), 3 / 4 ; 3 <i>idea that</i> group 3 and group 4 diverge more recently / AW or group 3 and group 4 on the same (evolutionary) branch ; differences: 4 first divergence resulted in, group 1, in, Figure 4.2 but in, Figure 4.1, group 1, formed later ; 5 group 2 are closely related to group, 3 / 4, in Figure 4.2 or group 2 are closely related to group 1 in, Figure 4.1 ;	3

Question	Answer	Marks
4(b)(ii)	any two from: 1 compare, base / nucleotide, sequences (of DNA) or identify, similarities / differences, in, base / nucleotide, sequences (of DNA) ; 2 more similarities means species, are more closely related / have a more recent common ancestor ; 3 ref. to arrange in groups of shared mutations ; 4 ref. bioinformatics ; A description	2
4(b)(iii)	data from more species ;	1
4(c)	any three from: 1 less genetic, diversity / variation ; 2 alleles not exchanged between non-sister chromatids ; 3 allele combinations remain the same (as parents) ; ora 4 linkage groups not broken / AW ; ora	3